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AIP Conf. Proc. 449, 502–512 (1998)

<https://doi.org/10.1063/1.56838>



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Overlay Metrology: The systematic, the random and the ugly

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Typical advanced lithographic processes require an overlay tolerance that is approximately 30% of the minimum feature size. To achieve this metrology must be limited to an error budget of 3-5% of the minimum feature size. A discussion of the general use of overlay data for assessment of stepper performance is followed by an overview of key overlay metrology equipment performance parameters (Tool Induced Shift - TIS) and the interaction with the process (Wafer Induced Shift - WIS). The interactions between the target, the process and the equipment will also be reviewed. Finally, use of this information in a case study of overlay target design for Tungsten Chemical Mechanical Planarization (W CMP) will be presented. Relevant process data will demonstrate how such targets can be used to effectively monitor both stepper (systematic error) and metrology tool (random error) performance.

INTRODUCTION

Semiconductor pattern overlay is the measure of vector displacement from one process level (substrate) to another level (resist), usually separated by an intermediary (thin film) layer. A top down view of a typical overlay measurement target is shown schematically in Figure 1.

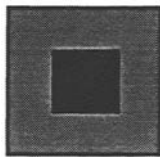


FIGURE 1. Overlay target schematic

Overlay error is defined as the planar distance from the center of the substrate target (gray box in figure 1) to the center of the resist defined target (black box in figure 1). Overlay measurement involves the determination of the centerline of each structure along both the X and Y axis. Centerline determination utilizes the symmetry around the structure center such that the error associated with edge determination will tend to cancel from each side of the structure. Conversely, in a linewidth determination the error associated with each edge combines in an additive fashion. Overlay measurement uncertainty is a complex function of tool, target and process interactions and is difficult to quantify. The device overlay tolerance specification is derived from the technology design rules. The circuit performance requirements are compared with established manufacturing tolerances. All error sources, from the mask to the measurement uncertainty are

combined either linearly (most conservative) or are added in quadrature (for Gaussian error distributions) to calculate the overlay tolerance. Successful process control of 0.5 micron process technology was possible through initial stepper matching and subsequent control of layer to layer overlay through the use of x y offsets. Process tolerances for the 0.5 micron generation, typically 150 nm on critical levels, were generous enough that the non-normal (multi-modal) behavior and dependence on location within a field did not affect the ability to meet target performance specifications. A typical example of the non-normal nature of the raw overlay measurement results is shown in Figure 2. Product misregistration overlay budgets for current 0.25 micron production

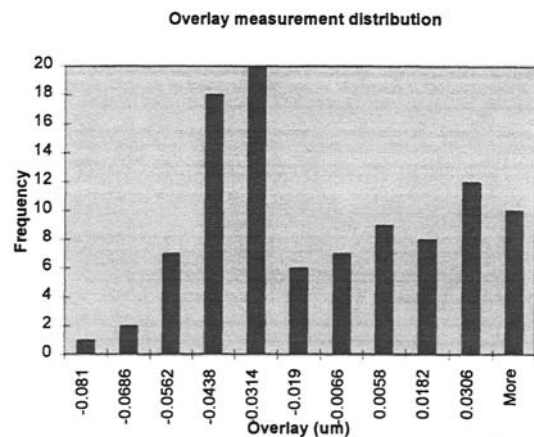


FIGURE 2. Non-Normal Overlay Distribution

process technologies however, are less than 100 nm. In order to consistently achieve these levels of performance every aspect of overlay error must be identified and

corrected. To that end, modeling is employed more frequently to identify, and where possible, correct the systematic sources of overlay error above and beyond the zero order offset terms. Moreover, process control is more difficult unless these systematic errors can be accounted for since their presence will distort the observed nature of the variation.

THE SYSTEMATIC:

The sources of overlay error attributable to the wafer stepper, generally arise from two sources; the lens (intrafield) and the stage (Grid). Intrafield errors can be further broken out into errors which are intrinsic to the optics of the stepper (distortions) and errors which are derived from interactions of the reticle with the stepper's optical subsystem. The former set of errors are characteristic of a given lens and are typically addressed during lens manufacture (direct control over a wafer stepper to control intrinsic lens distortions is not available). The residual lens distortions are accommodated for, at the price of individual stepper overlay performance, in the process of stepper matching. It is essential therefore, that each new lens be thoroughly characterized with respect to distortion since these errors are an integral part of the performance of the stepper in manufacturing.

Equation 1¹ is a mathematical model for all x direction intrafield error terms associated with steppers (a similar equation applies for the y direction).

$$(1) \delta x = \alpha x + (\delta M/M)x_0 - \Theta y_0 - t_1 x_0^2 - t_2 x_0 y_0 - E x_0 (x_0^2 + y_0^2) + F x_0 (x_0^2 + y_0^2)^2 + \text{Residuals}$$

From equation 1, the terms up to first order in x; Offset (α), Magnification ($\delta M/M$) and Rotation (Θ) can typically be controlled from the stepper. The higher order terms; Trapezoid (t_1 and t_2), third (E) and fifth (F) order distortions are either not easily controlled (trapezoid) or are intrinsic to the stepper lens (third and fifth order).

Stage derived errors may also be divided into those over which the process engineer has control and those which are "intrinsic" to the mechanical subsystem. As shown in equation 2², the systematic Grid errors are very similar in mathematical form to the intrafield or lens errors shown in equation 1 but unlike the intrafield case the terms apply across the entire wafer.

$$(2) \delta x = \alpha x + (\delta M_g/M_g)x_0 - \Theta_g y_0 + D x^2 + \text{Residuals}$$

$\delta M_g/M_g$ is the wafer scaling coefficient, Θ_g is the wafer rotation coefficient and D is the stage bow coefficient. Typically the stepper will allow direct control over only the scale and rotation terms. Accurate separation of the

various error components shown in equations 1 and 2 above is heavily dependent upon both sample plan and model statistics. The primary statistical assessment for goodness of fit is the 3 sigma coefficient error. This estimate provides a check of the statistical significance of the estimated coefficient against the other modeled systematic errors. Without this it is possible to accept error terms which are not significant and may add to the overall process variability. The residual error from the fit is also an indicator of the goodness of the model and will contain all random metrology error as well. In addition to assessing model performance, these error terms can be used to identify sample plan weaknesses. When establishing the metrology sample plan it is important to maximize symmetry and spatial coverage. These concepts derive from an understanding of the systematic error components due to the lithography tool. To achieve symmetry, die and intrafield site positions should be chosen such that they are balanced by an opposite die or site. For good spatial coverage, measurement locations must be positioned such that they adequately cover the area of interest (wafer or die) in both the x and y directions. Poor symmetry can lead to incorrect systematic error assessment. In the extreme intrafield case - one site per field, magnification error will appear as a translation error. Correction of this through an offset term will result in a 2x error at opposing die locations. Poor spatial coverage will also lead to incorrect systematic error assessment. For example; sampling the field in four corners only will result in exaggerated field magnification terms. This is due to the inclusion of (non-correctable) higher order (3rd and 5th) distortion terms. In this instance, adding one or two sites along the die edge (at X=max, Y=0 or X=0, Y=max) will allow for differentiation of the higher order systematic errors from magnification terms.

THE RANDOM:

Overlay measurement must be sensitive enough to discriminate error components derived from all sources. It is critical, therefore, that the contribution of the measurement tool to the total be minimized and well controlled for all process levels. Figure 3 shows relative tool performance versus lithographic error budget requirements for the last several process generations. The bold line in Figure 3 represents Overlay metrology specifications as outlined in the 1994 and 1997 SIA National Technology Roadmap for Semiconductors. The dashed line in Figure 3 represents worst case performance of overlay metrology equipment at Digital Semiconductor fabrication facilities over the same time period. As can be seen, the raw system performance has been and will continue to be adequate for several process generations.

However, overlay measurement difficulties arise in manufacturing from an interaction of the measurement equipment with the instance of the measurement feature (process variations and target design); primarily due to low edge contrast. Low contrast edges often result from advanced planarization techniques, such as Chemical Mechanical Polishing (CMP), used to accommodate a decreasing photolithography process window. These

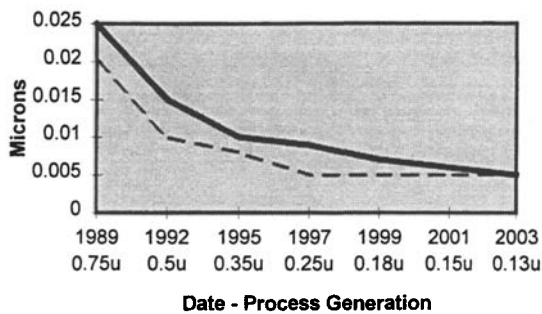


FIGURE 3. Performance of Overlay metrology equipment relative to SIA process error budget requirements.

problems require that implementation of overlay metrology be performed with a deeper understanding of the physics of the measurement instrument and the interaction of the process module with the overlay measurement target. The goal is to design effective targets that maximize edge contrast and allow equipment utilization in the most efficient modes of operation.

The inherent, tool limited, measurement accuracy is commonly referred to as Tool Induced Shift³ (TIS). TIS is quantified by measuring the same feature at 0 and 180 degree (wafer) rotations. TIS is one half the sum of the measurements from each orientation. The TIS error arises from; optical alignment, mode of illumination, and aberrations of the optics in the system. The second most common source of measurement error is termed Wafer Induced Shift (WIS). This error results primarily from physical changes to the measurement targets due to the manufacturing process (e.g. asymmetric film deposition / coat, severe noise due to grains or low contrast due to planarization). Unlike TIS, WIS is invariant with respect to rotation and, as a result is not as easy to identify. In many cases, it is not possible to quantify WIS until subsequent etch processing is performed. WIS must be addressed in both the measurement algorithms themselves and the overlay target design. The final component of the measurement contribution to the overlay error comes from the measurement precision of the tool. This component is easily quantified by performing repeated measurements of a sample, in a carefully designed Analysis of Variance (ANOVA) experiment. All of these

sources of measurement error combine to constitute the total measurement uncertainty.

Sources of TIS and contributions to WIS within the measurement system are primarily found in; the stage, the illumination system and the imaging optics. The stage must be well behaved in the X-Y plane during Z motion (where Z is perpendicular to the wafer surface). Any translations in X or Y which result from Z motion will contribute to measurement error. The worst case measurement scenario for this Z axis induced transitional error is during a dual focus measurement. In this measurement mode, which is used for structures exhibiting either topographies larger than two microns in Z or poor contrast, centerline estimates from the substrate and resist levels are taken at two separate focal planes. Any translation in X and Y which takes place during the move between focal planes will show up as measurement error (TIS).

The effect of optical aberrations on TIS must be examined from the perspective of centerline determination. Assuming that the target is located within the center of the microscopes field of view, the effect of all symmetric aberrations (e.g. spherical), will cancel. This is true for all focal positions since, in the centerline determination, the edge error will be symmetric (equal and opposite in value) across a centered measurement target.⁴ Asymmetric aberrations (e.g. coma), on the other hand, will result in an incorrect assessment of centerline for a symmetrically placed overlay measurement structure due to error asymmetry. Kirk⁴ has shown that brightfield optical systems, in the presence of 0.6λ of coma, can have a range of error in edge determination as large as 100 nm, over $1\ \mu\text{m}$ of defocus. From an overlay measurement perspective, the measurement instance most vulnerable to the effects of comatic aberrations is the single focus measurement on a dual layer sample; generally the standard measurement technique. In this case the two layers are at two different focus positions and each layer will experience different error magnitudes due to the comatic aberration.

Illumination will directly impact sensitivity to process variations and, if not properly aligned can result in large increases in TIS. The illumination bandwidth must be chosen such that it will; not expose photoresist, not be susceptible to diffraction phenomena, provide the required resolution, and maintain a stable output (intensity) over time. Typically broadband (450 - 650 nm) partially coherent⁵ light is set up in Kohler⁶ illumination in which the lamp filament is imaged on and completely fills the back focal plane of the objective. The aperture of the illuminator must be centered on the optical axis for symmetry. The importance of symmetric

(telecentric) illumination for the objective is derived from the centerline estimation process. If the illumination axis is not perpendicular to the wafer surface a lateral image shift with focus^{4,7,8} can result. Illumination can also be used to control system Numerical Aperture (NA) through coherence changes in the illumination subsystem. Initial work in this area⁹ was on targets for which process induced asymmetry was present in conjunction with a high level of noise (e.g. grainy metals). The results demonstrated that the choice of illumination wavelength most severely impacted those targets in which thin film effects were expected. Illumination alone was not shown to improve the accuracy of measurement results for asymmetric targets. However, by varying the system NA in this way¹⁰ it was shown that edge contrast enhancement is improved for certain grainy metal films. As always, trade-offs exist, and by increasing the system NA sensitivities to focus repeatability are accentuated.

Overlay measurement accuracy errors, if not recognized and accounted for can produce false systematic stepper errors. Zavecz¹¹ has reported that the introduction of systematic measurement errors into a simulated stepper set-up dataset results in the transposition of those errors to the stepper. TIS shows up in the translation term of the modeled stepper systematic errors and Pixel scale directly modifies the grid and field magnification (scale) coefficients. If these errors are not properly ascribed to the metrology tool they will end up as a stepper input correction which will further degrade product overlay performance. It is important to accurately quantify and assign the contribution of the measurement tool's error component to the measured value and to minimize its impact via a hardware modification (e.g. optical alignment / columnation) or software calibration.

THE UGLY:

TIS is observed to interact with process conditions, achieving different values for different; resist thickness', substrate surface roughness, substrate topographies, and structure designs.⁸ Figure 4 depicts the TIS performance of two different overlay measurement systems over six different process layers. The substrates shown include; metals, poly-silicon, field oxide and multilayer dielectric; the resist thickness' shown range from 0.5 μm to greater than 2 μm . The "Nominal" wafer is a single layer, (1 μm) resist on field oxide, with designed overlay offset, frame-in-frame, targets covering a 125 nm total offset range. Calculated TIS for this sample shows, minimum values for both measurement instruments relative to the other process levels. This is due to the planar nature of the structure which permits measurements at a single focal plane. If TIS were independent of the process effects, one

would expect to be able to predict a measurement instruments behavior on all process layers given an adequate understanding of the response in the ideal, minimum TIS instance. As can be seen from the figure, while it is generally true that minimum TIS results in best overall performance, it is also clear that there is a complex interaction between the various measurement schemes (optics, algorithms, illumination, etc.) represented by the different equipment and TIS performance on a given substrate. This data re-emphasizes the observations made during the discussion of the system optics TIS is a very complex function of; choice of focus method (e.g. single vs. Double grab), optical configuration, and substrate material. It is not possible to accurately predict the TIS response for a

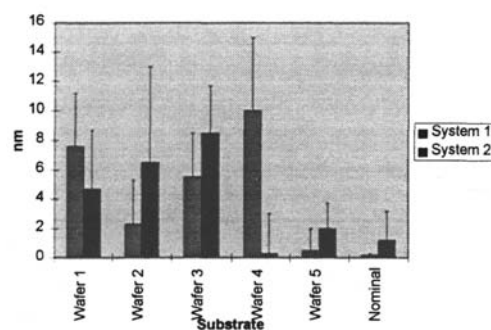


FIGURE 4. X TIS

given instrument on a new substrate based only upon an empirical understanding of the instruments performance on other materials. Figure 5 shows the interaction of focus and TIS. The data for each of the systems was collected in the single focus measurement mode using Wafer 3 from Figure 4 which has a resist thickness > 1.5 μm . The X axis of Figure 5 shows the deviation from best focus (0.0) in microns while the Y axis is the value

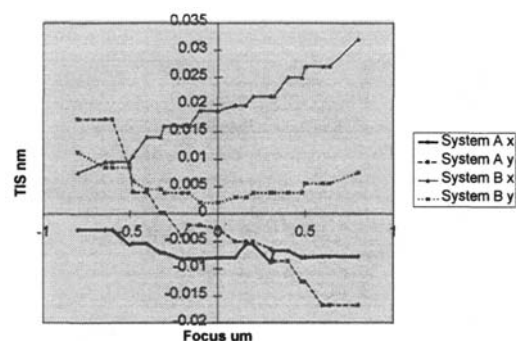


FIGURE 5. TIS vs. System focus

obtained for TIS. The behavior of the X TIS, for both systems, in the figure is typical of that which would be expected; a region of focus which corresponds to a stable operating point across the Depth of Focus (DOF) of the system optics. The Y TIS however is not so well behaved. Across the D.O.F. there does not appear to be a stable (TIS) point for either system. The anomalous behavior in Y TIS, corroborated by the two measurement tools, is believed to be due to an interaction with a target asymmetry.^{12,13} In fact, it is conceivable that the variations in TIS as shown in Figure 4, both within and across wafers, are due to an interaction of the tool asymmetry (TIS) and the process asymmetry (WIS). Even across a single wafer, process conditions are interacting with the fundamental tool setup in different ways at different wafer locations. This is the cause of the measurement errors of differing magnitudes shown by the error bars in Figure 4. As the hardware component of the overlay tools TIS has been reduced, process asymmetries have become a significant fraction of the observed measurement error. Target asymmetry is usually the result of proximity effects or asymmetric film deposition conditions. For example, a substrate step, in close proximity to the measurement target, can result in a locally thicker resist layer due to the planarizing nature of spin coating.¹⁴ There is also an effective increase in the exposure dose due to the greater reflectivity in the vicinity of the step. This effective dose change, coupled with the thicker resist across the step, results in an asymmetric target. Measurements taken on structures exhibiting this asymmetry by Starikov et al¹² demonstrate that a mean overlay difference of 70 nm, along the axis affected by the asymmetry was observed as compared to a structure which was not influenced by proximity effects. Further simulations by Starikov show that the introduction of even a 0.5% resist slope can result in a 23 nm measurement error. Tanaka¹⁵ reports of a target asymmetry which results from uneven deposition of a film. In the instance cited a metal sputter deposition process produces film thickness' which are dependent upon topography and orientation with respect to the incident sputtered atoms. This results in a "snowdrift" effect within the box-in-box overlay target, whereby the metal is piled up in one side of the overlay target. From an overlay measurement perspective, the "snowdrift" effect results in an inaccurate measurement due to an incorrect estimate of the "true" target centerline. The measurement inaccuracy which results from this type of target asymmetry can be greater than 100 nm (3 sigma) when comparing measurements taken following etch processing with standard overlay measurements taken following the develop process.¹⁵ The WIS component of inaccuracy must be addressed through measurement target design optimization and measurement algorithm development.

Modern overlay measurement algorithms are moving beyond the threshold algorithm. This departure is due to accuracy difficulties which result from the complex film stacks used in state of the art processes. Given the lack of symmetry within some targets, basic threshold algorithms are prone to systematic measurement error. Determination of a specified threshold on the image waveform, corresponding to a point on the edge of the overlay target will not match reality. In addition, asymmetry as a result of tool (e.g. minor illumination misalignment) or mark sources (grainy film or image contrast variation) causes different edge detection errors for the left and right edges. Errors greater than 50 nm can be expected to result. It is essential that more than a simple interpretation of the signal waveform for a percentage of the threshold be performed. The measurement algorithm can take advantage of "a priori" knowledge of the overlay target¹³ and the physics of the subsequent interactions with the films and resist layers. All measurement algorithms, in one way or another, compare the centerline (or the intersection of two orthogonal centerlines) of a substrate layer to the photoresist layer. Certain "a priori" facts are known about the "ideal" target which can be used to interpret the manifestation of that target for a given substrate and target design. Use of this information in the measurement process can lead to improved measurement accuracy, even in the absence of certified standards. Starikov et al¹³ describe an algorithm which takes advantage of the inherent symmetry and redundancies present in overlay target design. Since target designs follow standard dimension practices, such information can be used to assess any asymmetries resulting exclusively from tool or process. Another algorithm which has been proposed to address process variations is a modified intercorrelation algorithm.¹⁶ This algorithm uses all of the data contained within a defined region encompassing the actual edge, rather than just a single threshold point along the maximum slope. Such a method provides added accuracy since using the entire waveform tends to average out any low contrast or noisy segments of the waveform. The intercorrelation data can be also used to judge the symmetry of the target. In a well-formed highly symmetric target, the output of the intercorrelation is expected to be well behaved. Based upon this, it is conceivable that a library of intercorrelation responses to varied target designs over all substrate and resist combinations could be used for target design optimization.

Another avenue used to address WIS is overlay target design. An example of typical overlay target design is shown in Figure 6. Typical target dimensions are for the inner box - a = 1-2 μm b = 6 - 11 μm , c = 10 - 15 μm and for the outer box: A = 1 - 4 μm , B = 16 - 26 μm C = 20 -

30 μm . (e.g. the classic box-in box structure of Fig 2 would have dimensions $a=5$, $b=10$, $c=10$ $A=10$, $B=20$, $C=20$). A number of possible combinations of segmented frame outer, solid box inner frame can be considered in relation to the amount of information derived from the structure, for a specific process layer. When evaluating target designs, it is also important to study different phases of the inner and outer boxes. Substrate trench structures will provide, in many instances vastly different contrast than mesa structures. In addition, the choice of contrast “tone” of the structure (e.g. resist inner box / substrate outer box vs. substrate inner / resist outer) will produce different measurement results. Typically the

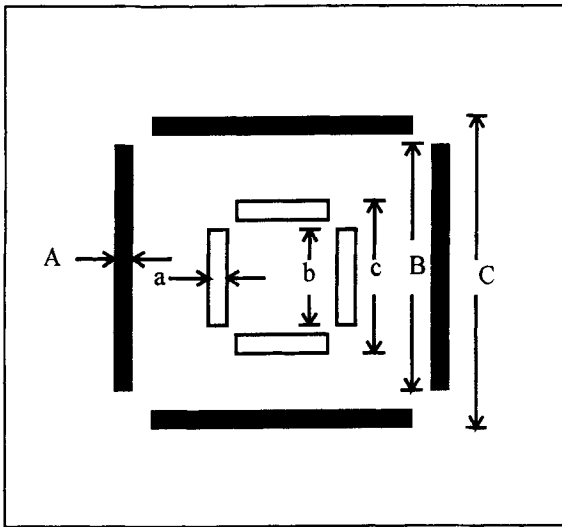


FIGURE 6. Typical automated overlay structure design

resist box provides well defined edges with little asymmetry (in the absence of proximity effects) whereas the substrate defined box is subject to process variations (grainy materials, reflectivity variations etc.). The best structures tend to be those which provide the greatest amount of redundant information for processing by the measurement algorithm. Although a detailed analysis of the target-to-process interaction must be performed for each process it is possible to define several generic overlay target design rules:

1. Make substrate level the outer “Box”
2. Maximize edge area for lowest contrast / highest noise edge
3. Use multiple edges for substrate and resist features
4. Select target profile to minimize z distance between levels.
5. Don’t obscure substrate edges with resist overcoat
6. Strive to make targets comparable to reticle layout (e.g. contact and via levels % open area)

7. Avoid proximity effects near substrate edges (keep adjacent edges $> 5 \mu\text{m}$ from target edges).

OVERLAY METROLOGY CASE STUDY: W CMP

As process technology approached the 0.35 μm generation, the use of Chemical Mechanical Polishing (CMP) became more pervasive. The increase in the number and thickness of layers being manufactured on the silicon wafer has resulted in large variations in the planarity of the surface. Such non-planarity’s can often consume the entire Depth-of-Focus of the photolithography equipment. Further, such a high degree of non-planarity has been shown to impact device yield due to contamination following dry etch and poor step coverage of subsequent metallization layers. Initially CMP was relegated to polishing of interlayer dielectric materials (ILDs)¹⁷ but as expertise in CMP grew, its application to other process modules followed. CMP has been implemented for shallow trench isolation; for polysilicon where improved planarization has resulted in reduced CD variation(2) and for metallizations (plug and inlaid) where reduced defectivity and improved recess control has been achieved. While very beneficial for process performance, CMP has resulted in extremely low contrast overlay (and alignment) targets which have significantly impacted metrology performance.

As the name implies, CMP is a process by which a sacrificial Oxide, metal, or polysilicon layer is removed through a combination of chemical and mechanical action. The wafer is placed on a mechanical carrier, circuit side down into a slurry (alkaline for oxide CMP and acidic for metal CMP). The carrier applies both downforce and rotational motion to the wafer which sits on a pad resting on the polishing table which is also in motion (rotational or linear) with respect to the carrier. The CMP process itself is very difficult to control. Planarization is a function of both wafer attributes (e.g. feature size, circuit density, and film type) and a host of CMP equipment parameters (time, downforce, slurry pH and particle size, pad hardness, pad condition, table rpm, wafer rpm, and etc.) many of which change over time. Additionally, the thickness uniformity of the planarized layer can vary significantly across both die and wafer. From an overlay measurement perspective this means that target behavior will vary across the wafer as well as from wafer-wafer and lot-lot. In oxide CMP, the material removal is heavily dependent on the mechanical aspects of the polishing. The goal of the oxide CMP process is to remove or minimize topography (perfect planarity) while maintaining uniformity of thickness for all affected films across a 200 mm wafer. Unfortunately for oxide CMP

these two goals are mutually exclusive. This is in some ways fortunate for the overlay metrology function. The goal of overlay metrology target design (in direct opposition to the goal of the planarization process) is to maintain sufficient edge contrast (i.e. topography) for adequate measurement performance. The fact that perfect planarization can only be achieved at the (very high) cost of across-wafer non-uniformity, results in the possibility that some feature steps will remain. The design of the overlay measurement targets must take this fact into account and utilize additional understanding of the dependencies of planarization to achieve this end.

For Tungsten (W) CMP the slurry chemistry plays a more significant role than it does in dielectric CMP, enabling fairly high selectivity to oxide (as high as 150:1).¹⁸ Additionally, due to grain size differences, different thickness of as deposited W will have different polish rates¹⁹ further enhancing oxide selectivity. In theory, it should be possible, in this situation, to adequately control the CMP process to achieve the desired level of W removal. In reality however, the selectivity to oxide is greatly reduced as a function of pattern density¹⁸. This phenomena, which is dependent on all aspects of the W CMP process (esp. pad hardness, slurry type, and thickness of as deposited W), is termed erosion and is shown schematically in figure 7 (right).



FIGURE 7. Dishing (left) and erosion (right) ¹⁸

The dark areas in Figure 7 represent the oxide, the light areas in the figure represent the W. The impact of excessive erosion on the overlay target will be to completely planarize the edge step, rendering the target

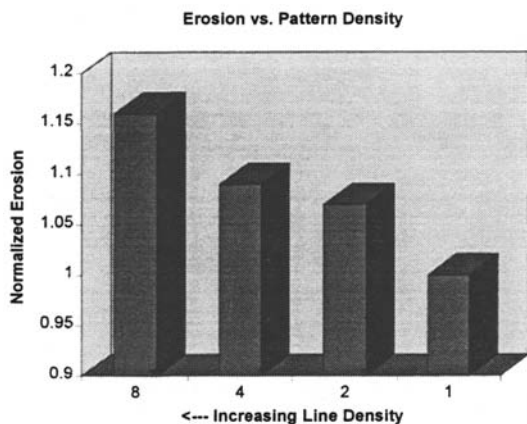


FIGURE 8: Measured Erosion as a Function of Pattern Density ¹⁸

unmeasurable. This occurs because of higher polishing forces on the oxide edges. Since these dense areas clear first, the oxide between the W plugs is exposed to the polishing process for a longer duration. Higher local pressure exists in this case as well since less of the wafer oxide surface area is in contact with the polishing pad in these dense regions as compared to regions of large unpatterned oxide planes. Figure 8¹⁸ depicts the magnitude of erosion to be expected for increasing line density. It is clear that for overlay target design, with respect to erosion, a relatively isolated feature would be expected to have minimal erosion vs. the rest of the die.

The other circuit dependent CMP phenomena is called dishing and is a function of feature width. This is shown schematically in figure 7 left. In this case, once the W has been removed from unpatterned areas (dark areas in figure 7) polishing of oxide will continue but at a much slower rate. However W patterned areas will continue to polish and those areas which are large enough for the pad to deflect into the W feature will produce non-uniform polishing from the center-out. Figure 9¹⁸ depicts the magnitude of dishing to be expected as a function of increasing linewidth. For overlay target design, it is clear that the trenchwidth of the substrate box must not exceed a critical value (dependent upon the CMP pad hardness) in order to avoid dishing effects.

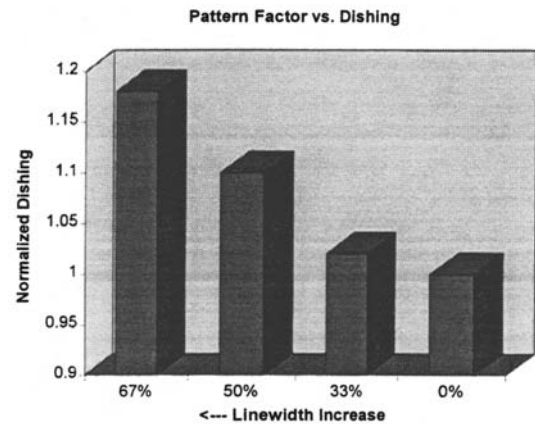


FIGURE 9. Measured Dishing as a Function of Linewidth. ¹⁸

When excessive dishing occurs in an overlay measurement feature, the edge is effectively smoothed over the entire width of the trench (as opposed to having a clean sharp step transition from oxide to W). In summary, for overlay target design, the impact to edge contrast from planarity, uniformity, erosion and dishing must be given careful consideration. In some instances these phenomena can be used to promote edge contrast. Careful quantification of the dependence of erosion and dishing on circuit density and feature size (see figures 8

and 9), typically done during the development phase for the CMP process can be used to define design guidelines for overlay metrology targets.

As has been discussed, the primary effect of W CMP on the overlay targets is to significantly reduce the edge contrast by reducing and rounding the step height of the features used to define the edge. Prior to W CMP, typical edge definition for a standard W etchback process was on the order of 2000 Angstroms (measured by AFM to be the stepheight from oxide surround to the W surface). With W CMP this step is typically reduced to 200 Angstroms or smaller. This small step is then covered with metallization prior to the photo patterning. Such a feature poses significant challenges to the overlay measurement equipment. A treatment of the theoretical constraints of overlay measurement with respect to precise and accurate results has been performed by A. Shchemelinin et al²⁰ While the mathematics are beyond the scope of the present work, the general observations and conclusions are worth noting and will be related to system and target improvements for W CMP overlay. These workers found that: the measurement error is proportional to the square root of the noise. In this case the noise refers to all sources present in the measurement activity (electronic or Johnson noise and target noise). Another interesting finding, which supports the use of multiple edges, states that the measurement error increases with targets which contain intervals of constant signal (as would be found in the box-in-box target design). Thus increasing the number of edges and reducing their separation will result in improved measurement precision. Further support for the use of multiple edges was found in this analysis where the measurement precision is reduced by a factor between k and the square root of k (where k is equal to the number of lines present in the measurement target). Finally this theoretical analysis of measurement errors found that the measurement error is inversely proportional to the system resolution. Any system modifications which improve resolution (e.g. raise NA, lower illumination wavelength, use confocal or interferometric imaging) will improve measurement precision. Based upon considerations such as these, overlay equipment manufacturers have attempted to address the issue of low contrast edges through a number of novel approaches. These include; the use of phase information, Lens NA optimization, illumination optimization, algorithm optimization and signal processing / spatial filtering. Fundamental system design for W CMP overlay metrology examines the entire system as a whole.²¹ In general, short wavelength illumination (for improved resolution) and a choice of bandwidth which minimizes both chromatic aberrations

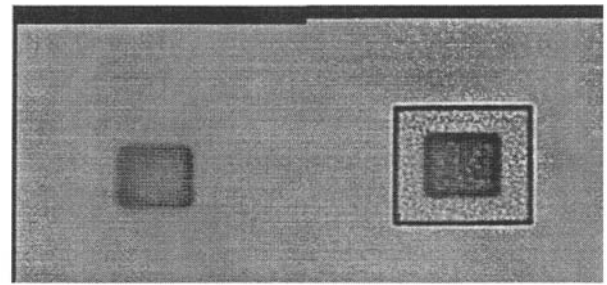


FIGURE 10. Contrast Improvements due to Image Enhancement 22

and thin film effects²¹. Resolution, as discussed, contributes significantly to the performance of measurement algorithms. In many cases it is possible to exceed theoretical limits through spatial oversampling and Fourier based algorithms. However, for CMP, contrast not resolution, is the critical system parameter. Contrast enhancement can be accomplished by lowering the system NA. This is achieved through increasing the system response to lower spatial frequency elements by modifying the illumination towards coherence.²¹ Significant contrast enhancement can be obtained with minimum impact to resolution ($> 20\%$ reduction). The danger with this approach comes as a result of increased diffraction effects. Finally elimination of all system induced noise (e.g. Johnson noise) must be accomplished to the point that the image noise is due solely to sample noise (e.g. background limited). Examples of the improvements from such system optimization are shown in Figure 10²² (brightfield-left, enhanced-right).

A second system approach involves utilization of phase information. The optical phase change upon reflection from light waves traveling from the substrate as compared to those from the edge area is quite large and can be used to enhance the edge contrast. The commercial implementation of phase imaging²³ utilizes a Linnik interferometer to obtain the phase information from the overlay target. The Linnik interferometer utilizes an incoherent, broadband light source and relies on interference from the sample light path and the

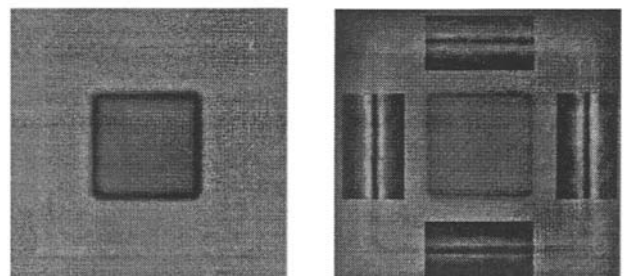


FIGURE 11. Contrast Improvements due to Phase Imaging 24

EXPERIMENTAL:

reference light path when the sample is within the coherence region. The resulting signal is produced from very small region in z (perpendicular to the plane of the sample) which is defined by the coherence length of the optics. As the sample is scanned in z , a coherence probe image (or "Cloud Plot" - refer to Figure 13) is constructed from the interference fringes within the coherence region. Contained within this image is phase information which provides the basis for the measurement algorithms. Figure 11 shows the comparison of a brightfield image (left) and a phase enhanced Coherence Probe image (right) of a typical, low contrast target at W CMP.

Measurement algorithms are another area where the overlay equipment has been significantly improved to accommodate the low contrast targets produced by CMP. One of the most straightforward improvements involves utilizing a priori knowledge of the overlay target (e.g. edge size and redundancy characteristics). This is an extension of some of the algorithmic methods discussed earlier. In practice the image enhancement attempts to utilize components known to be present in the image which the "noise factors" do not share. One such method²⁵ is to use the linear characteristics of the square shape of the typical overlay measurement target to enhance the image numerically. An example of this enhancement is shown in Figure 12²⁶ (brightfield-left, enhanced-right). Clearly the image contrast is greatly enhanced using this method. Comparison between stan-

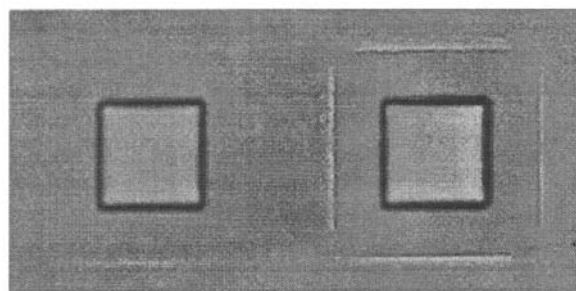


FIGURE 12. Contrast Improvements due to a priori Target data²⁶

dard brightfield measurements and the image enhanced measurements²⁵ has demonstrated significant improvements in precision. There was also an accompanying improvement in measurement accuracy, identified using stepper modeling. These same workers have also examined reducing the wavelength of the illumination and increasing the objective lens NA. This serves to reduce the depth of focus and improve the vertical resolution of the measurement instrument making it more sensitive to the low contrast edges of the CMP target.

AFM Characterization of the effect of the W etchback process on the overlay measurement targets (from figure 5: $a=5\text{ }\mu\text{m}$, $b=10$, $c=10$; $A=1$, $B=16$, $C=20$ - resist box in substrate bars), which preceded the W CMP process, gave a step height into the substrate bar of $> 2000\text{ }\text{\AA}$. As the process was moved to W CMP, initially no modifications were done to the overlay targets. A dramatic degradation of the lot-lot variability of overlay measurement results was observed. AFM characterization of the W CMP targets showed a substrate edge step of $200 - 400\text{ }\text{\AA}$. An attempt to utilize phase-based image enhancement techniques demonstrated success in reducing the measurement variability at the expense of throughput and ease of measurement setup. Figure 13 shows the brightfield image of the W CMP target. Very little edge contrast is available for measurement. Figure 14 shows a coherence probe generated cross-sectional image of the overlay target. The step height at the substrate edge, highlighted by the arrow, is quite small. As can be seen the phase enhancement technique can be quite effective in increasing the edge contrast. Short loop characterization of the W etchback process using overlay measurement performance and AFM step height characterization showed that a severe degradation in observed measurement variability occurred for step heights less than $1000\text{ }\text{\AA}$. An examination of the design variables available for the overlay target, in conjunction with available CMP characterization data led to target design

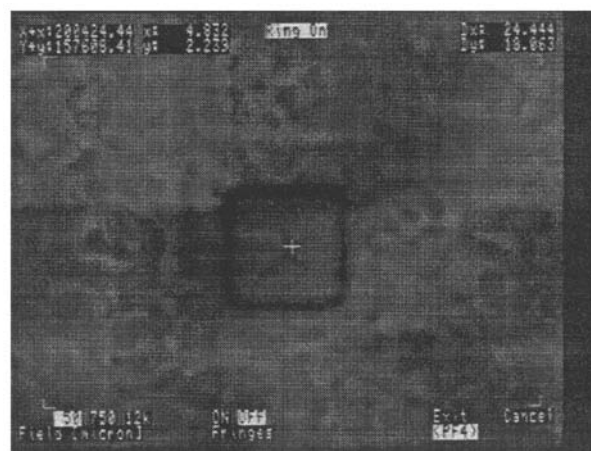


FIGURE 13. Brightfield View of W CMP Overlay Target.

modifications. For the experimental work: The substrate bar width (A in figure 5): $A=1, 2, 4\text{ }\mu\text{m}$; The CMP parameters: slurry type, downforce, and overpolish; W thickness at two levels; and trench depth variations (by

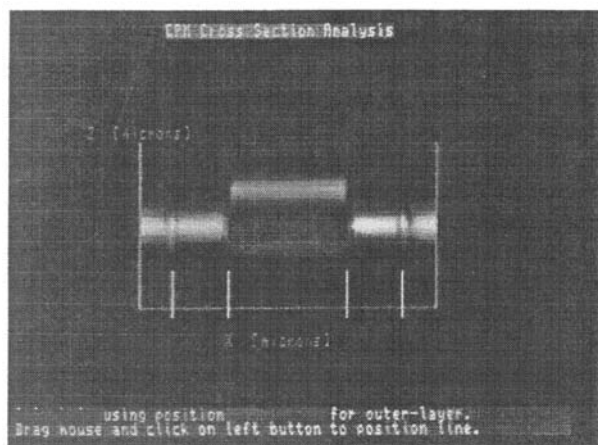


FIGURE 14. "Cloud Plot" of W CMP Overlay Target.

removing etch stops) were examined for isolated target structures. The primary output: measurement performance was supported by AFM characterization of the step height for each of the samples.

RESULTS:

Targets in which the W didn't completely fill the trench worked best. In the case of this experiment that includes targets of $A = 2 \mu\text{m}$ and $4 \mu\text{m}$. However edge contrast was lost in the larger targets due, for the most part, to

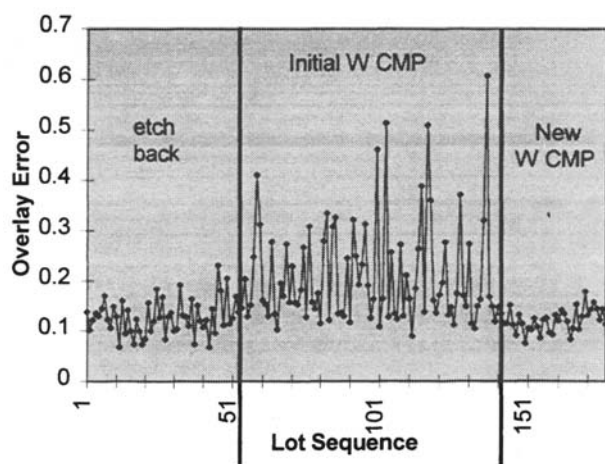


FIGURE 15. Manufacturing Data Demonstrating Overlay Target Performance.

dishing which smoothed the edge over the $4 \mu\text{m}$ width, creating considerable uncertainty in the edge location for the metrology tool. From a CMP perspective; downforce had less impact on the metrology than overpolish and

reduced W deposition created the best contrast targets. The $2 \mu\text{m}$ target with the etchstop removed was designed into all product and test reticles. Figure 15 shows the resulting improvement on product material, due to the combination of the target modifications, the W thickness optimization and the trench depth increase. An improvement of nearly 3x from the "old" target with W CMP (Lot 52 - Lot 145) and the "new" target W CMP (Lot 146 - Lot 160) can be seen. Some small improvement is also visible from "new" W CMP as compared to the W etchback (Lot 1 - Lot 51).

DISCUSSION:

From the results achieved in the preceding study, in conjunction with some recent work by S. Hsu and co-workers at National Semiconductor Corp.,²⁷ it is now possible to articulate some overlay measurement target design rules for W CMP. In addition to the general rules listed earlier the following rules are proposed:

- 1) Substrate bar formed by isolated space to avoid erosion.
- 2) Maximize edge definition by maximizing post W CMP Step height:
 - a) Minimized deposited W thickness
 - b) Optimize trench width to 6 - 8 times minimum feature size (Max width determined by dishing).
 - c) maximize trench depth (remove etch stops).

Work continues in this area to develop a single design rule which would account for the effects of erosion and dishing on the overlay target as a function of; target Critical Dimension, pre-W trench depth and W film thickness.

CONCLUSIONS:

Data has been presented which demonstrates the capability of current generation overlay metrology equipment for sub-0.25 micron process technologies, assuming moderate contrast overlay targets. It has also been shown that for certain process modules, most notably W CMP, serious effort must be devoted in the target design to maintain viable overlay metrology. Detailed examination of target response to CMP was used to design an experiment. From this, several broad design guidelines, both for generic overlay measurement targets and for target design specific to W CMP have been related. Improvements in overlay metrology have been shown to result, in an actual production application, from the use of these design guidelines.

As overlay metrology advances beyond the current focus on stable target design the next level of refinement with respect to CMP and its affect on these measurements must be undertaken. It has already been demonstrated that²⁸ physical changes to the substrate target (in a way which is analogous to the "snowdrift effect" discussed earlier) can be detected through the use of scanned probe measurements. These workers have discovered that an induced rotation, which is not the result of the lithography process, shows up in a systematic analysis of the target asymmetry.

ACKNOWLEDGEMENTS:

The authors wish to acknowledge Frank Krupa for his CMP expertise, his patience and his critical review of this work.

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